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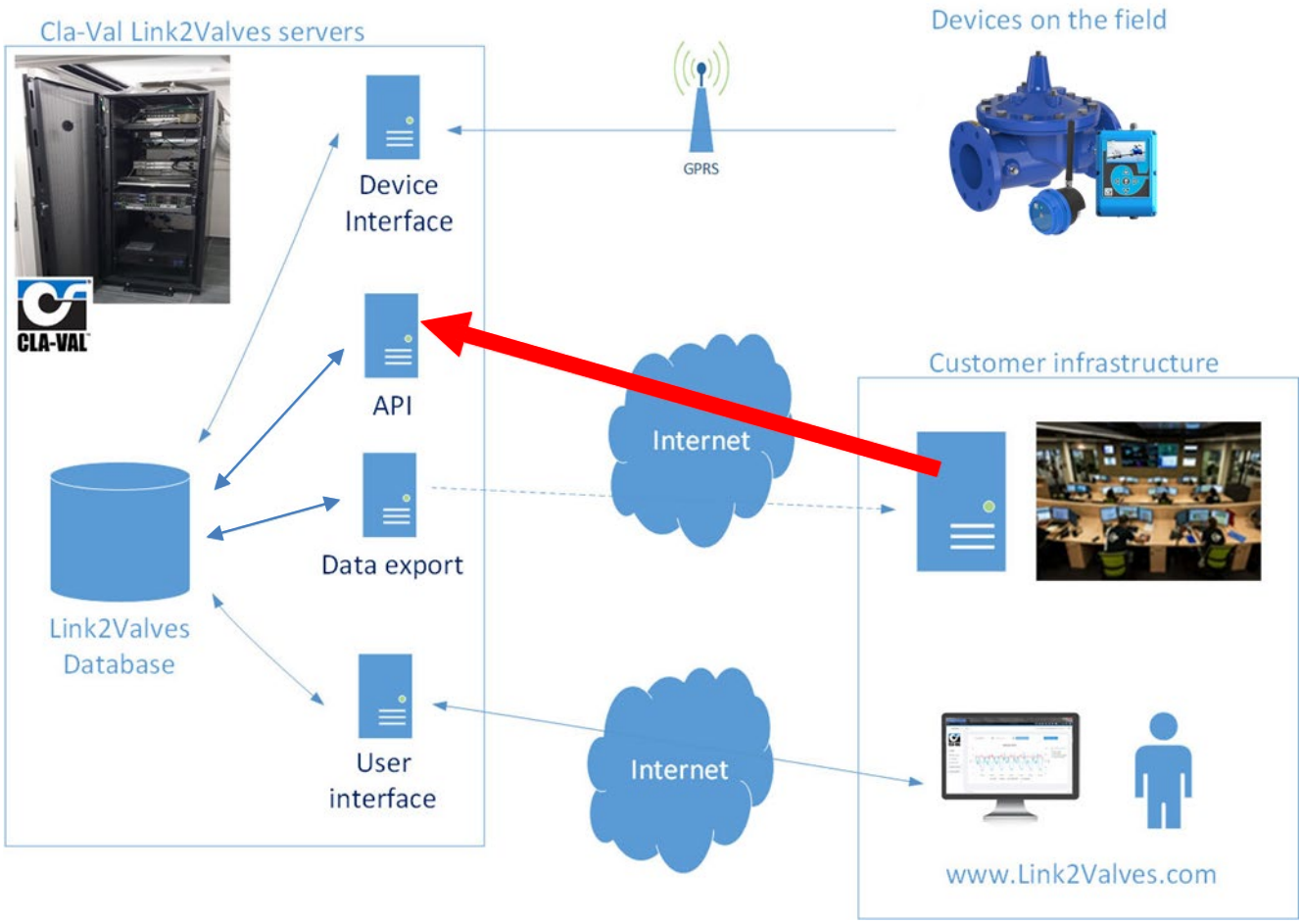
1 OVERVIEW

Link2Valves' JSON API allows customer IT infrastructure to retrieve asset management data such as valve information from the Link2Valves web platform.

This way, Link2Valves data can be exploited by the customer infrastructure without the need to manually visit the Link2Valves website.

The approach involves making requests using HTTPS and authorized through a client certificate, which enforces strong, state-of-the-art Internet security standards.

An indication of the role of the public API inside the Link2Valves ecosystem is shown by the red arrow in the figure below:



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2 ADVANTAGES

- Bypass the GUI of the web platform for quick retrieval of all asset data
- Easy parsing of the responses by leveraging the standard JSON format
- Secure, seamless integration into existing IT infrastructure by leveraging the ubiquitous HTTPS protocol
- Requests being outgoing only; no need for the customer to open their IT environment to receive any kind of connection by CLA-VAL

3 REQUIREMENTS

3.1 INFRASTRUCTURE REQUIREMENTS

The customer is responsible of developing the business logic that:

- Will regularly launch the HTTP requests, integrating the client certificate generated by CLA-VAL.
- Will interpret the JSON result.
- Will save the result, back it up, process or otherwise consume the data inside the customer's infrastructure.

The customer will also need to follow a simple procedure to generate its own private client certificate. CLA-VAL will provide assistance during the process.

3.2 SECURITY REQUIREMENTS

Requests must use HTTPS.

Authorization is achieved through the combination of the following:

- The client certificate provided in the request.
- A Link2Valves user, created by CLA-VAL on request, associated with the provided client certificate.
- The assignment to the Link2Valves user of permissions allowing access to the requested devices.

CLA-VAL can, on request, provide assistance to put in place any of these authorization requirements.

4 REQUESTS

4.1 GET ASSETS

https://www.link2valves.com/public/api/assets

Purpose: Retrieve the details of all assets stored in Link2Valves that are accessible by the client certificate used in the request.

Method: GET

Response: A Customer or SiteOrg JSON object, depending on the access level of the client certificate.

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5 OBJECT DESCRIPTION

CUSTOMER		
Information for a customer. Can contain site organizations and sites.		
Property	**Type**	**Description**
temporaryId	Integer	Internal use only.
siteOrgs	Array of Site Org objects	Structure used to contain and organize sites.
sites	Array of Site objects	Structure used to contain and organize valves.
id	Integer	Id of the customer.
parentId	Integer	Id of the parent container holding the customer.
parentType	Integer	Type of the parent container holding the customer.
name	String	Name of the customer.
streetAddr	String	Street address of the customer.
city	String	City of the customer.
state	String	State/province of the customer.
country	String	Country of the customer.
zip	String	Postal code of the customer.
repName	String	First name of the representative from the customer.
repSurname	String	Surname of the representative from the customer.
repPhoneNum	String	Phone number of the representative from the customer.
repEmail	String	Email address of the representative from the customer.
repJobTitle	String	Job title of the representative from the customer.
notes	String	Miscellaneous notes for the customer.
geolocationDisabled	Boolean	If true, all geolocation functions will be disabled for the customer.

SITE ORG		
Information for a site organization. Can contain site organizations and sites.		
Property	**Type**	**Description**
temporaryId	Integer	Internal use only.
siteOrgs	Array of SiteOrg objects	Structure used to contain and organize sites.
sites	Array of Site objects	Structure used to contain and organize valves.
id	Integer	Id of the site organization.
parentId	Integer	Id of the parent container of the site organization.
parentType	Integer	Type of the parent container of the site organization.
name	String	Name of the site organization.
streetAddr	String	Street address of the site organization.
city	String	City of the site organization.
state	String	State/province of the site organization.
country	String	Country of the site organization.
zip	String	Postal code of the site organization.

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repName	String	First name of the representative of the site organization.
repSurname	String	Surname of the representative of the site organization.
repPhoneNum	String	Phone number of the representative of the site organization.
repEmail	String	Email address of the representative of the site organization.
repJobTitle	String	Job title of the representative of the site organization.
notes	String	Miscellaneous notes for the site organization.
geolocationDisabled	Boolean	If true, all geolocation functions will be disabled for the site organization.

SITE		
Information for a site. Can contain valves, photos, attachments, and custom fields.		
Property	**Type**	**Description**
temporaryId	Integer	Internal use only.
parentId	Integer	Id of the parent container of the site.
parentType	Integer	Type of the parent container of the site.
confinedSpace	Boolean	True if the site is considered a confined space.
twoPeople	Boolean	True if two people must be present to access the site.
files	Array of File objects	Used internally. Will always be an empty array.
photos	Array of Photo objects	Photos attached to the site.
attachments	Array of Attachment objects	Attachments attached to the site.
valves	Array of Valve objects	Valves contained in the site.
id	Integer	Id of the site.
name	String	Name of the site.
addressField	String	Address of the site.
latitude	Floating point	Latitude coordinate of the site.
longitude	Floating point	Longitude coordinate of the site.
customFields	Array of CustomField objects	Custom key-value pairs associated with the site.
worstMaintenanceStatus	String	Shows the maintenance status of the valve which is most in need of maintenance at the site.

VALVE		
Information for a valve. Can contain pilots, maintenance records, photos, attachments, tags, and custom fields.		
Property	**Type**	**Description**
parentId	Integer	Id of the parent container of the site.
temporaryId	Integer	Internal use only.
modelNumber	ModelNumber object	Model number associated with the valve.
dateCode	String	Date code associated with the valve.
portType	String	“Full” or “Reduced”.
bodyType	String	“Globe” or “Angle”.
material	String	Material of the valve body.

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endConnections	String	“Flanged”, “Grooved”, or “Threaded”.
trim	String	“Standard”, “KO”, or “KOL”.
tubingFittingMat	String	Tubing and fitting material for the valve pilot system.
inletPrs	String	Expected inlet pressure on the valve during operation.
outletPrs	String	Expected outlet pressure on the valve during operation.
pressureRating	String	“ANSI 150” or “ANSI 300”.
installDate	String	Install date for the valve (ISO 8601).
pilots	Array of Pilot objects	Pilot valves installed on the main valve.
files	Array of File objects	Used internally. Will always be an empty array.
photos	Array of Photo objects	Photos attached to the site.
attachments	Array of Attachment objects	Attachments attached to the site.
records	Array of Record objects	Maintenance records for the valve.
maintenanceStatus	String	Maintenance status of the valve.
fluidType	String	Fluid type used in the valve.
maintenanceWarnThresholdMonths	Integer	Number of months before the maintenance due date that the valve will be marked as needing maintenance.
maintenanceWarnThresholdDays	Integer	Number of days before the maintenance due date that the valve will be marked as needing maintenance.
rubberKitRec	Floating point	Number of years before rubber kit is recommended, rounded to nearest quarter (default 3 years).
rubberKitRecTimeUnit	String	“year”
repairKitRec	Floating point	Number of years before repair kit is recommended, rounded to nearest quarter (default 6 years).
repairKitRecTimeUnit	String	“year”
rebuildKitRec	Floating point	Number of years before rebuild kit is recommended, rounded to nearest quarter (default 10 years).
rebuildKitRecTimeUnit	String	“year”
nextVisitDate	String	Date of the next recommended maintenance activity (ISO 8601).
lastHealthyDate	String	The most recent date that the maintnenanceStatus was “ok” (ISO 8601).
returnEvery	Integer	Number of years between recommended valve inspections.
timeUnit	String	“year"
id	Integer	Id of the valve.
name	String	Name of the valve.
brand	String	Name of the valve manufacturer.
partNumber	String	Part number of the valve.
intModelNumber	String	International model number of the valve.
size	String	Size of the valve.
tags	Array of strings	Custom tags associated with the valve for filtering at the site-level.
notes	String	Miscellaneous notes for the valve.

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customFields	Array of CustomField objects	Custom key-value pairs associated with the valve.
installed	Integer	Unix time of the valve installation.

MODEL NUMBER		
Information for a valve model number.		
Property	**Type**	**Description**
prefix	String	Prefix of the valve model number.
suffix	String	Suffix of the valve model number.
optional	String	Options appended to the end of a valve model number.

PILOT		
Information for a valve pilot installed on a main valve.		
Property	**Type**	**Description**
model	String	Pilot valve model.
setPointNote	String	Setpoint associated with the pilot valve.

ATTACHMENT		
Information for an attachment.		
Property	**Type**	**Description**
contents	String	Base64 encoding of the binary file data
mediaType	String	File format of the attachment, e.g. “application/pdf”.
id	Integer	Id of the attachment.
name	String	Name of the attachment.
type	String	File format of the attachment (duplicate field).
size	Integer	Size of the attachment in bytes.
category	String	“Attachment”

PHOTO		
Information for a photo.		
Property	**Type**	**Description**
contents	String	Base64 encoding of the binary file data
mediaType	String	File format of the photo, e.g. “image/jpeg”.
id	Integer	Id of the photo.
name	String	Name of the photo.
type	String	File format of the photo (duplicate field).
size	Integer	Size of the photo in bytes.
category	String	Category of the photo, e.g. “Photo_Overview”.

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RECORD		
Information for a valve maintenance record.		
Property	**Type**	**Description**
id	Integer	Id of the record.
parentId	Integer	Id of the valve containing the record.
temporaryId	Integer	Used internally.
date	String	Date of the activities in the record (ISO 8601).
title	String	Title of the record.
hasAttachments	Boolean	True if the record has attachments.
reasonForVisit	String	Reason for the activities in the record.
satisfiedPM	Boolean	If true, the valve's maintenanceStatus will be set to "ok" as of the date of the record.
laborTimeHours	Floating point	Number of labor hours associated with the activities in the record.
notes	String	Miscellaneous notes associated with the record.

CUSTOM FIELD		
Information for a custom field.		
Property	**Type**	**Description**
name	String	Name of the custom field.
value	String	Value of the custom field.

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